

OSTROLEKA, Poland

WMO: 122850

Lat: 53.066N

Lon: 21.534E

Elev: 308

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.9	(c) 9.7	(d) -1.0	(e) 5.3	(f) 6.1	(g) 3.9	(h) 6.8	(i) 11.4	(j) 19.7	(k) 42.2	(l) 17.7	(m) 39.5	(n) 4.1	(o) 70	(p) 0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	86.1	68.8	82.9	66.9	80.2	65.6	71.1	82.0	69.0	79.7	67.2	76.8	6.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.4	101.7	75.9	65.4	95.1	73.4	63.6	89.1	71.5	35.1	81.9	33.3	79.7	31.8	76.9	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.7	(b) 13.8	(c) 12.1	(e) -1.7	(f) 93.0	(g) 8.4	(h) 2.9	(i) -7.7	(j) 95.1	(k) -12.6	(l) 96.8	(m) -17.3	(n) 98.5	(o) -23.3	(p) 100.6		
(a) 15.7	(b) 13.8	(c) 12.1	(e) -0.3	(f) 74.4	(g) 9.6	(h) 1.7	(i) -7.2	(j) 75.7	(k) -12.8	(l) 76.7	(m) -18.2	(n) 77.7	(o) -25.2	(p) 78.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	28.1	30.3	37.1	47.2	56.7	62.1	66.2	65.2	56.5	46.9	39.2	33.2
	DBStd	15.22	9.55	9.78	8.22	7.33	7.72	6.49	5.94	5.97	6.05	7.30	6.70	8.39
	HDD50	2808	677	552	403	137	27	1	0	0	14	147	329	521
	HDD65	6633	1142	972	865	534	275	130	59	73	262	560	775	986
	CDD50	1893	0	0	2	53	234	365	502	472	209	52	4	0
	CDD65	242	0	0	0	0	17	43	95	79	8	0	0	0
	CDH74	2612	0	0	0	16	230	538	914	818	96	0	0	0
	CDH80	685	0	0	0	1	29	133	250	252	20	0	0	0
Wind	WSAvg	5.9	6.9	6.5	6.7	5.9	5.5	5.5	5.3	4.9	5.1	5.7	6.4	6.7
Precipitation	PrecAvg	24.8	1.5	1.2	1.6	1.9	2.4	3.2	3.3	2.8	2.6	1.9	1.7	1.5
	PrecMax	42.6	4.3	2.3	4.8	3.8	5.0	8.3	8.6	9.8	7.5	4.6	3.4	2.8
	PrecMin	18.0	0.2	0.2	0.3	0.2	1.0	1.0	0.7	0.4	0.4	0.2	0.2	0.4
	PrecStd	5.8	0.9	0.5	1.0	0.9	0.9	1.8	1.8	1.9	2.0	1.1	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	50.6	62.1	75.4	83.6	89.5	90.4	92.0	82.5	70.3	57.6	51.1
		MCWB	44.7	44.9	50.2	58.7	64.6	73.0	68.6	69.0	66.4	60.3	53.4	48.0
	2%	DB	45.3	47.6	57.3	70.0	79.9	83.4	85.8	85.8	76.4	65.9	53.4	48.3
		MCWB	42.8	44.0	46.9	55.3	63.4	67.2	68.2	68.5	63.5	57.9	50.2	45.9
	5%	DB	42.9	44.6	53.0	65.8	76.2	80.3	82.6	82.1	72.1	61.6	50.5	45.9
		MCWB	40.8	41.5	44.9	52.5	61.4	65.7	67.1	66.7	61.3	55.9	47.9	43.7
	10%	DB	40.2	41.9	49.2	61.7	72.0	76.7	79.3	78.3	68.6	58.2	48.3	43.4
		MCWB	38.4	39.2	43.5	50.3	59.0	63.8	65.8	65.5	59.2	53.9	46.1	41.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	46.4	51.8	60.8	67.4	76.5	73.4	72.9	67.7	61.8	54.2	49.1
		MCDB	47.0	48.9	59.5	72.6	79.0	83.8	83.6	85.8	79.4	68.2	57.2	50.5
	2%	WB	43.2	44.3	48.5	56.6	65.1	70.6	70.6	70.5	64.9	59.1	50.4	46.1
		MCDB	45.2	47.1	54.4	67.4	76.2	80.9	81.2	82.6	73.8	63.9	52.7	47.9
	5%	WB	41.0	41.7	46.2	54.2	63.0	67.3	68.9	68.4	62.5	56.7	48.4	44.1
		MCDB	43.0	44.5	51.4	63.6	73.5	76.6	79.0	78.8	69.7	61.0	50.2	45.7
	10%	WB	38.5	39.3	43.7	51.7	60.7	64.7	67.0	66.6	60.5	54.3	46.3	41.9
		MCDB	40.2	41.7	48.8	58.8	69.8	73.3	76.4	75.7	67.1	57.9	48.0	43.5
Mean Daily Temperature Range	5% DB	MDBR	7.9	10.2	14.9	20.2	21.4	21.0	19.8	20.9	18.9	15.0	8.4	7.2
		MCDBR	8.9	12.5	22.2	27.1	26.4	26.5	24.4	25.8	24.8	20.6	11.8	8.5
		MCWBR	7.6	9.5	13.8	15.0	12.2	11.7	9.9	11.0	12.8	13.1	9.0	7.2
	5% WB	MCDBR	8.6	12.1	19.6	24.8	22.6	22.7	21.2	23.0	21.8	18.7	11.2	8.3
MCWBR		7.4	9.5	13.3	14.4	11.0	11.8	9.5	10.6	11.9	12.6	8.9	7.3	
Clear-Sky Solar Irradiance	taub	0.308	0.339	0.373	0.420	0.407	0.403	0.436	0.430	0.388	0.362	0.351	0.307	
	taud	2.218	2.220	2.229	2.161	2.260	2.300	2.226	2.237	2.327	2.380	2.286	2.262	
	Ebn at Noon	206	237	254	256	267	269	257	251	248	229	188	184	
	Edh at Noon	21	29	35	42	40	39	42	39	31	24	20	17	
All-Sky Solar Radiation	RadAvg	207	431	821	1305	1663	1744	1621	1405	994	535	210	141	
	RadStd	39	72	126	173	163	154	183	125	142	91	22	18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-277	+148	N/A

Nomenclature: See separate page